

Solution Requirements

Date	06 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely log in using a username and password before accessing the system.</i>	High
2	Authorization levels	<i>Students can access emotion detection and learning recommendations. Administrators can manage users, datasets, and reports.</i>	High
3	External interfaces	<i>The system integrates with a webcam, Streamlit interface, OpenCV, TensorFlow/Keras emotion detection model, and educational content resources.</i>	High

4	Transactions processing	<i>The system captures facial images, processes emotions in real time, generates recommendations, and stores learning progress.</i>	High
5	Reporting	<i>Generates emotion analysis results, learning recommendations, student progress reports, and performance summaries.</i>	Medium
6	Business rules, etc.	<i>Learning resources are recommended based on the detected emotion. Emotion detection must complete before recommendations are displayed.</i>	High
7	Compliance to laws or regulations	<i>Student information is processed securely. Facial images are used only during the active session unless explicitly saved with user consent.</i>	High
8	Other	<i>Supports real-time emotion detection, personalized learning assistance, and progress tracking.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>Detect emotions and generate recommendations quickly.</i>	<i>Response time < 3 seconds</i>
2	Scalability	<i>Support increasing numbers of students and additional learning modules.</i>	<i>Supports 1000+ users</i>
3	Security & Data Privacy	<i>Secure user credentials and protect emotion-related data.</i>	<i>Encrypted data transmission and secure authentication</i>
4	Reliability & Availability	<i>System should remain operational with minimal downtime.</i>	<i>99% uptime</i>
5	Usability & Accessibility	<i>Simple, intuitive Streamlit interface accessible to students with minimal training.</i>	<i>Easy navigation with clear UI</i>
6	Other	<i>Maintainable, portable, and easy to update with new emotion models and subjects.</i>	<i>Modular architecture with straightforward deployment</i>